BlueSky Social Network Analysis of User Behavior and Community Dynamics

1. Introduction

Social networks have become an integral part of modern communication and information exchange, providing unique opportunities to analyze user behavior, interaction patterns, and community dynamics. This report explores the structure and dynamics of the BlueSky social network, focusing on interactions among users through likes and follows. By employing advanced network analysis methods, this study aims to uncover key insights into user influence, community structure, connectivity, and gender-based interactions.

The dataset, derived from BlueSky's platform via API integration, represents a directed network where nodes signify individual users and edges indicate their interactions. Advanced tools like Gephi are utilized for visualizing and analyzing centrality measures, clustering, and modularity to derive actionable insights. This comprehensive analysis contributes to a deeper understanding of social behavior, information flow, and structural properties within the BlueSky network, laying the groundwork for future research and strategic interventions.

2. Data Collection Process

Authentication and API Integration:

The data collection process commenced with authenticating the BlueSky API using a unique handle and password. Successful authentication generated an access token, which granted secure access to user-generated data, including posts, likes, and follower relationships.

Steps in Data Collection

- **Posts Retrieval:** The fetch_feed function was used to retrieve up to 800 posts in batches, leveraging pagination to ensure comprehensive data extraction. Real-time progress updates allowed efficient monitoring of the process.
- **Likes and Follows:**

- Likes: The fetch_likes function identified users who liked a post, creating a direct connection between the liker and the post's author.
- Followers: Using the fetch_followers function, the followers of each post author were recorded, adding another layer of interaction to the dataset.

Gender Prediction:

The gender of users was inferred using an external Gender Prediction API. Display names and labels were cleaned to remove special characters, and the first name was extracted for gender analysis. A caching mechanism was employed to minimize redundant API calls, ensuring efficient and accurate predictions.

Data Preparation:

The collected data was processed into two key datasets for analysis:

- **Nodes Dataset:** Each node represents a unique user, identified by their handle. Additional attributes include the display name, label, inferred gender, and profile URL for reference.
- **Edges Dataset:** Each edge represents an interaction between two users, categorized as either "like" or "follows." This categorization enables the differentiation of interaction types within the network.

3. Exported Datasets for Analysis

The processed data was carefully structured and exported into two CSV files for utilization in Gephi, facilitating detailed network analysis and visualization:

- **nodes.csv:** This dataset captures user-specific attributes, including a unique identifier (ID), display name, inferred gender, and profile URL. These attributes enable demographic and behavioral analysis while providing cross-referencing capabilities with the original network.
- **edges.csv:** This dataset outlines the interactions between users, detailing the source (initiator), target (receiver), and type of interaction (either "like" or "follow"). This distinction is crucial for understanding the nature of relationships within the network.

These datasets form the basis for network visualization and advanced analytical tasks, including the computation of centrality measures, community detection, modularity, and clustering coefficients. They also provide the framework for visualizing the network's topology, revealing patterns of influence, interaction, and community formation. By structuring the data in this manner, the analysis achieves a high degree of accuracy and relevance, enabling meaningful exploration of the BlueSky social network.

4. Visualizations and Analysis

The visualization of the BlueSky social network, comprising 18,781 nodes and 22,414 directed edges, provides a comprehensive depiction of user interactions through likes and follows. This directed graph effectively illustrates the flow of interactions and highlights how users engage within the network.

Network Composition and Interaction Dynamics:

The nodes, representing individual users, are visually scaled based on degree centrality to emphasize users with significant connections. Node colors reflect gender, offering demographic insights:

- **Males**: Represented in blue, comprising 55.59% of the network.
- **Females**: Depicted in pink, accounting for 28.22%.
- **Unknown**: Shown in black, representing 16%.

Edges are distinctly colored to differentiate between the two primary interaction types:

- **Follows**: Red edges, representing a slightly higher proportion of connections at 50.59%.
- **Likes**: Green edges, comprising 49.41%.

The visualization reveals a densely populated core of the network, indicating a high degree of interaction among central users. In contrast, the periphery consists of sparsely connected nodes, reflecting users with fewer interactions. This contrast highlights the network's structural balance, with influential users at the core and less engaged users at the edges.

The strategic application of color coding for nodes and edges enables a quick and intuitive understanding of gender demographics and interaction patterns. Additionally, scaling node

sizes based on connectivity underscores the influence of highly connected individuals within the network.

This graphical representation, as shown in Figure 1, serves as a foundational element of the analysis. It not only visualizes the network's structure but also sets the stage for further exploration into centrality measures, clustering coefficients, modularity, and community structures.

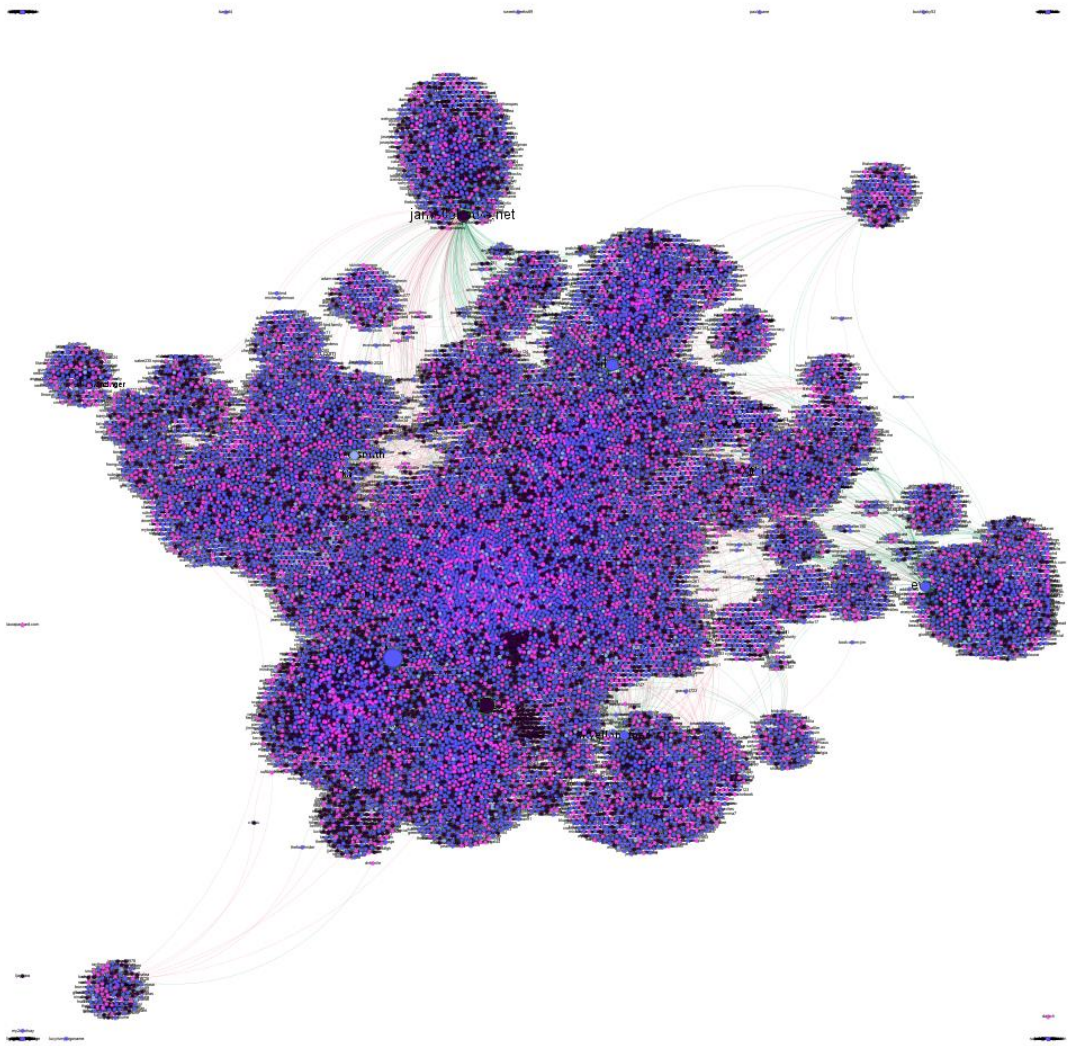


Figure 1: Graphical representation of the BlueSky social network, illustrating gender-based interactions and connectivity dynamics

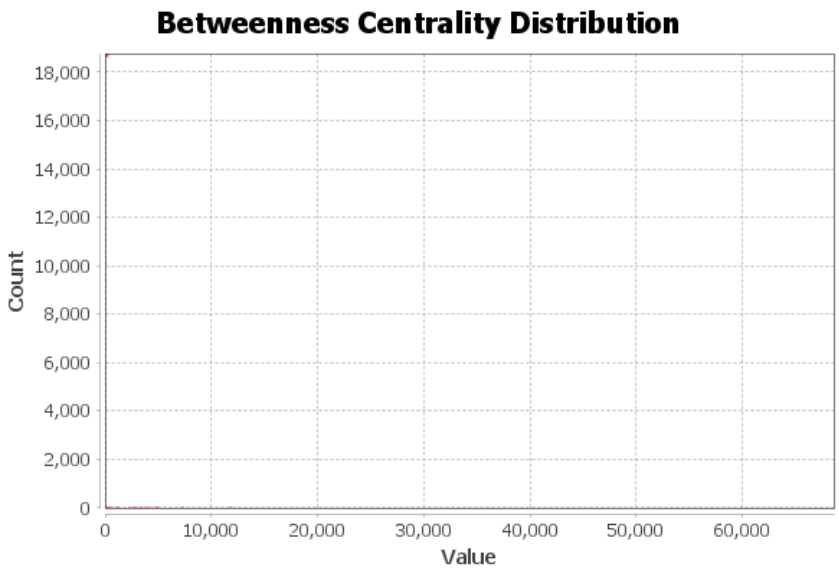
5. Centrality Measures and Graph Topology

The analysis of centrality measures and graph distance metrics offers critical insights into the BlueSky social network's structure, influence distribution, and overall connectivity.

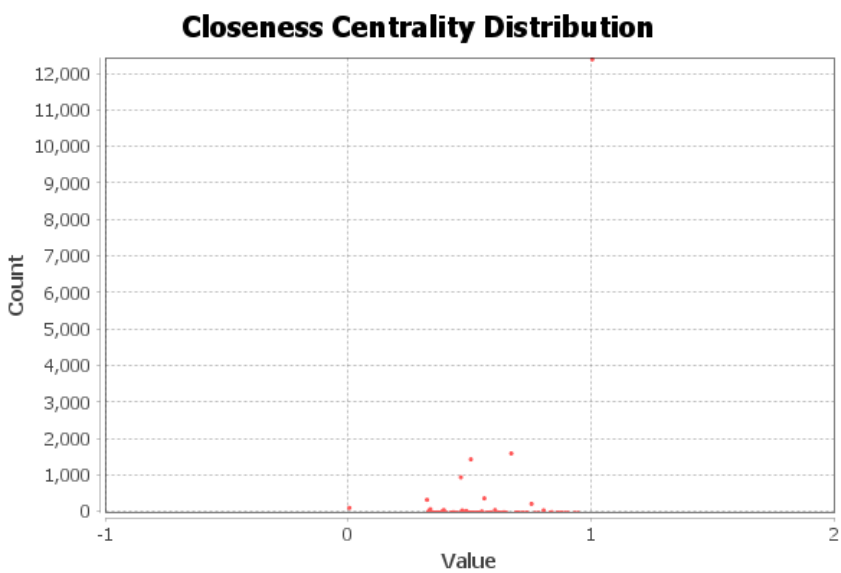
The findings highlight the roles of influential nodes, the network's reachability, and its small-world characteristics.

Centrality Measures

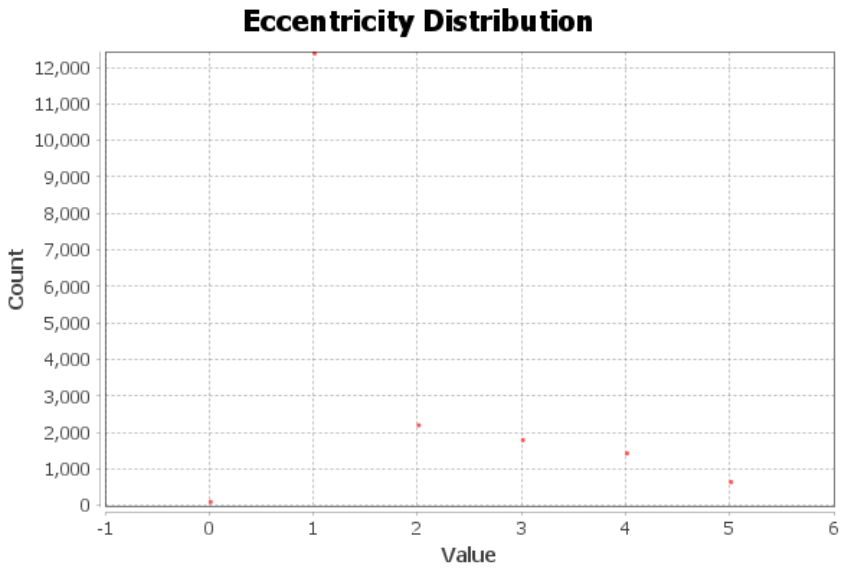
- **Betweenness Centrality**: The majority of nodes have betweenness values near zero, indicating limited roles as intermediaries. However, a select few nodes with high betweenness act as strategic bridges, facilitating communication between otherwise disconnected clusters.



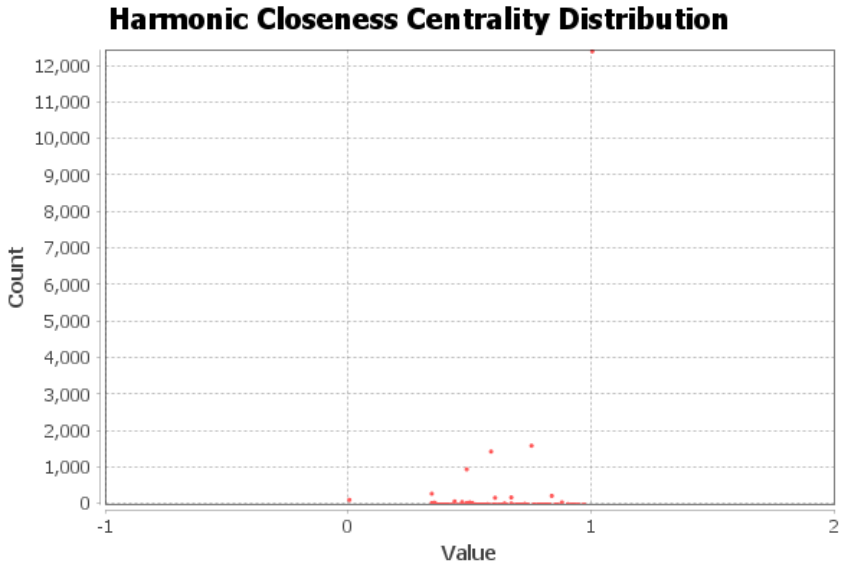
- **Closeness Centrality**: The values are skewed towards the lower range, reflecting that most nodes are farther from others in terms of shortest paths. This suggests a hierarchical or clustered network structure.



- **Eccentricity**: The eccentricity distribution ranges from 0 to 5, with most nodes concentrated near the lower end. This indicates a relatively compact network where nodes are generally accessible from others within a limited number of steps.



- **Harmonic Closeness Centrality**: Similar to closeness centrality, harmonic closeness centrality highlights that while a few nodes are strategically placed for easy accessibility, the majority of nodes are less connected and farther from the network's core.



Graph Distance Metrics

- **Diameter**: The network's diameter is 5, indicating the longest shortest path between any two nodes. This suggests a tightly interconnected structure.

- **Average Path Length**: The average path length is 2.15, emphasizing the small-world nature of the network, where most nodes are connected through relatively few intermediaries.

- **Radius**: A radius of 0 suggests the presence of outlier nodes disconnected from the main network, requiring further investigation.

As depicted in Figure 2, the visualization integrates these findings using the ForceAtlas 2 layout. Nodes are scaled based on closeness centrality to highlight accessibility, colored to reflect betweenness centrality and emphasize key intermediaries, and connected with edge thickness corresponding to edge betweenness. This comprehensive depiction underscores the network's influential nodes and critical connections, providing a clear representation of its structure and topology.

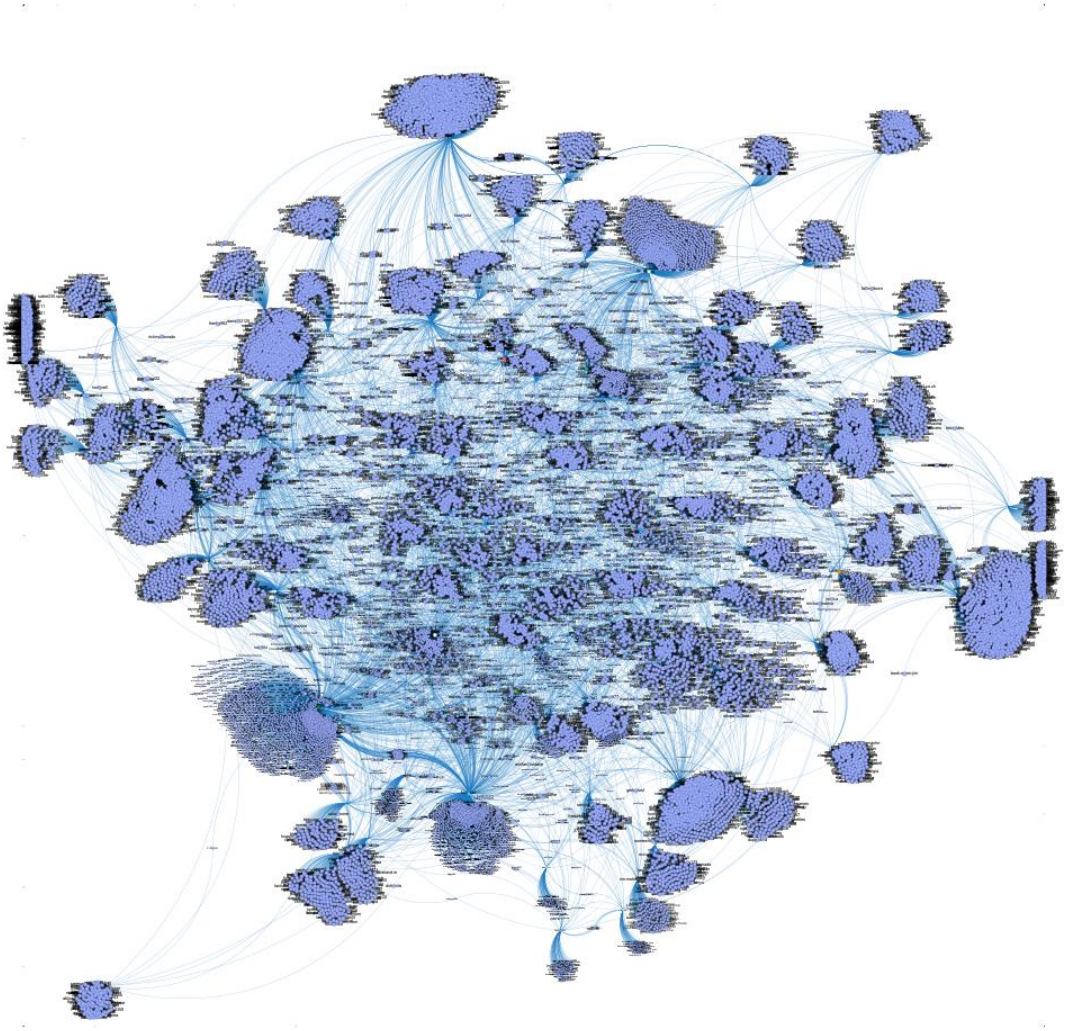


Figure 2: Visualization of centrality measures and network topology, showcasing key nodes and connections

6. Component Measures Analysis

The analysis of connected components within the directed BlueSky network sheds light on its structural clustering and the dominance of the giant component. The findings emphasize the network's cohesion, highlighting critical clusters and isolated structures.

- **Weakly Connected Components:** The network comprises 61 weakly connected components, representing clusters of nodes indirectly connected when edge direction is ignored.

- **Strongly Connected Components:** A total of 18,773 strongly connected components were identified, indicating that many nodes or small clusters are isolated when edge directionality is considered.

Key Observations:

1. **Giant Component Dominance**
    - A significant 99.68% of the nodes belong to the largest connected component (highlighted in blue), underscoring the centralized and cohesive nature of the network.
    - The remaining nodes are distributed across smaller, isolated components forming peripheral clusters.
2. **Component Visualization**
    - Nodes were partitioned by Component ID, with those in the largest component prominently colored blue.
    - Node sizes reflect their Strongly Connected Component ID, visually emphasizing key clusters and isolated nodes.
3. **Interpretation**
    - The giant component represents the network's core, where most interactions and connections occur, driving overall cohesion.
    - Smaller components signify isolated users or sub-networks, disconnected from the main structure and less influential in network dynamics.

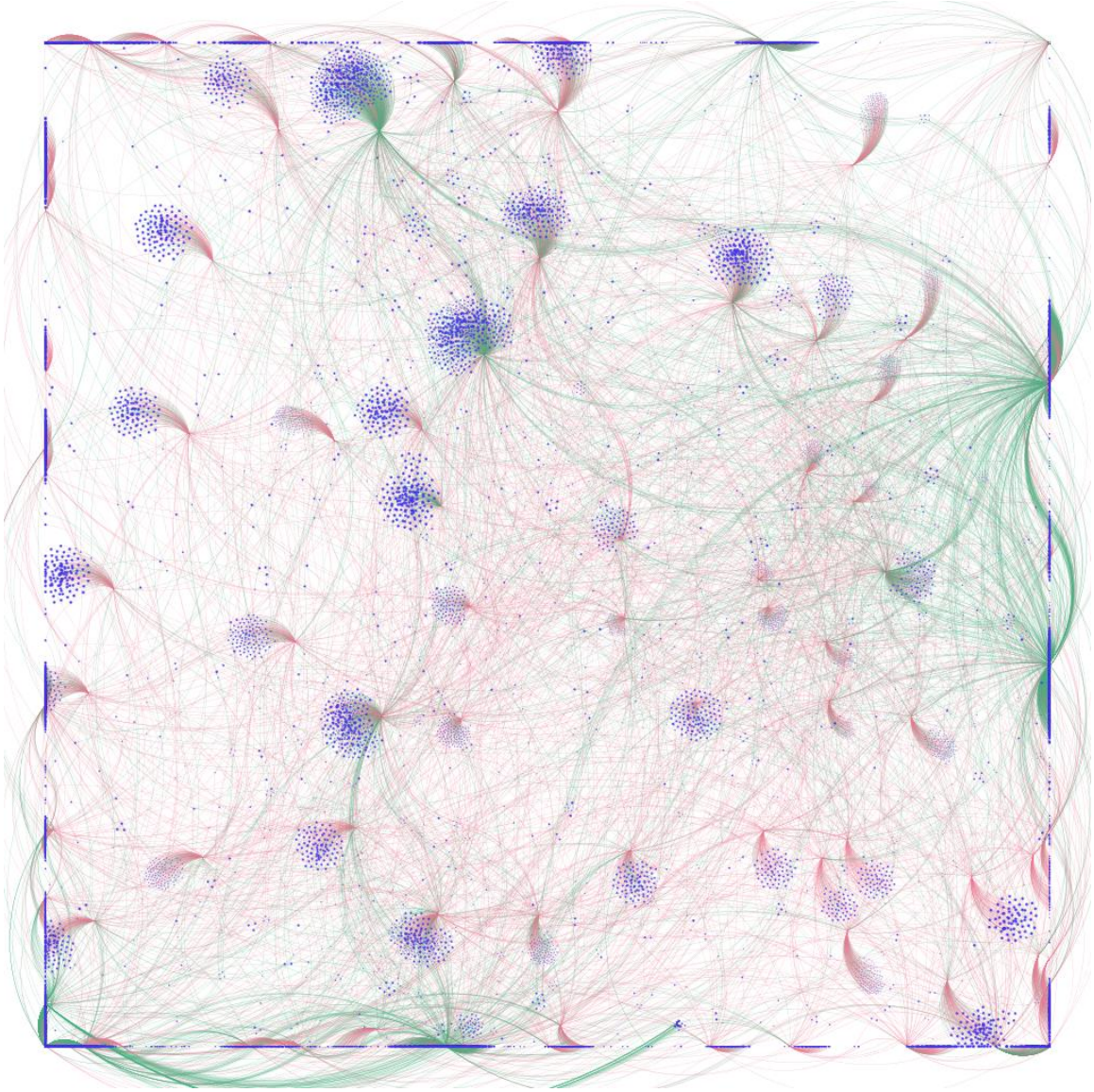


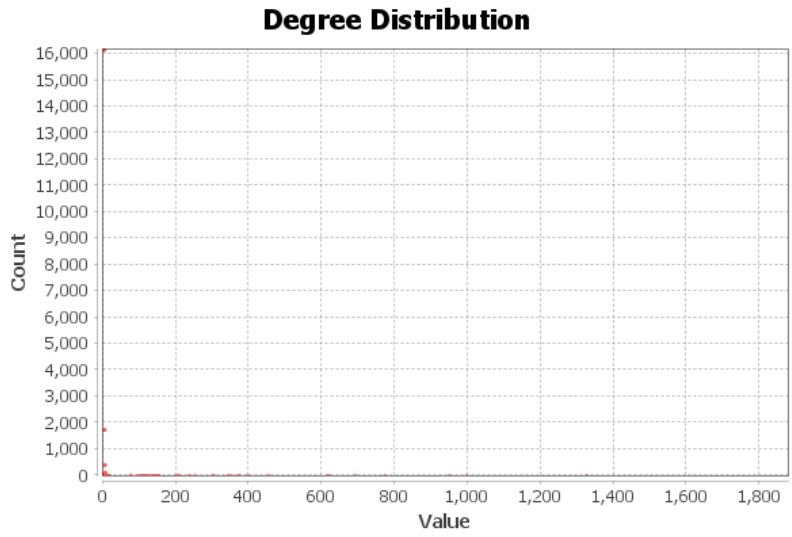
Figure 3: Visualization of connected components, emphasizing the giant component and isolated clusters

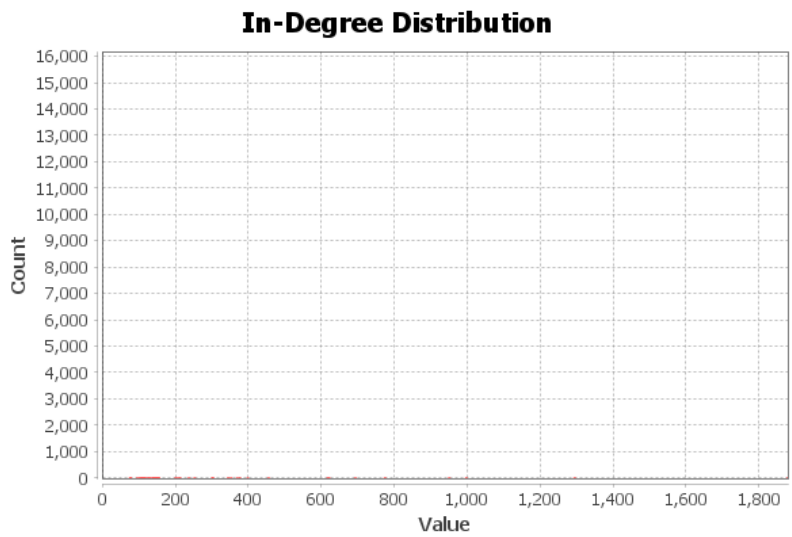
The visualization, as depicted in Figure 3, effectively highlights these structural features, offering critical insights into the network's hierarchical organization and its degree of centralization. This understanding is essential for exploring interaction dynamics and influence within the BlueSky platform.

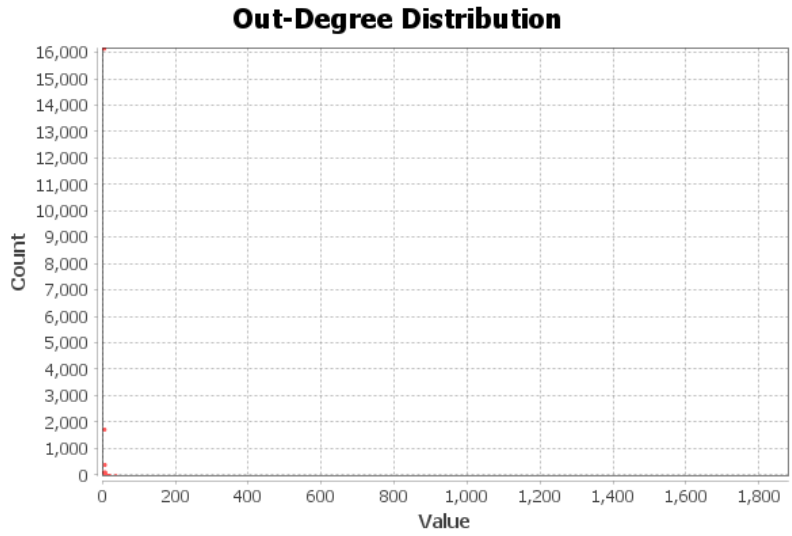
7. Centrality Measures

Centrality measures are crucial for identifying influential nodes and understanding their roles in the network's structure and dynamics. These metrics shed light on how nodes contribute to information flow, connectivity, and overall network resilience. Below is an analysis of the key centrality measures computed:

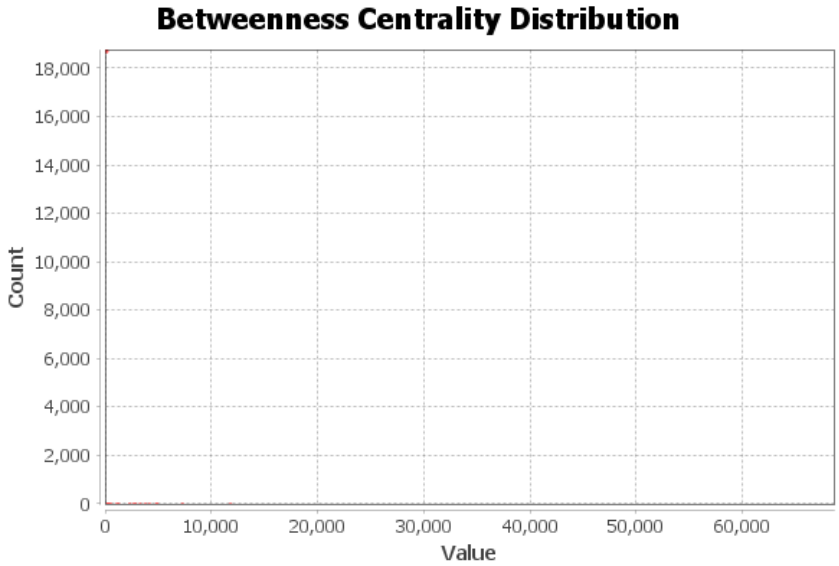
- **Degree Centrality:** Degree centrality identifies the most connected nodes in the network. With an average degree of 1.193, the results indicate active yet limited participation in interactions, as the majority of nodes maintain few direct connections.



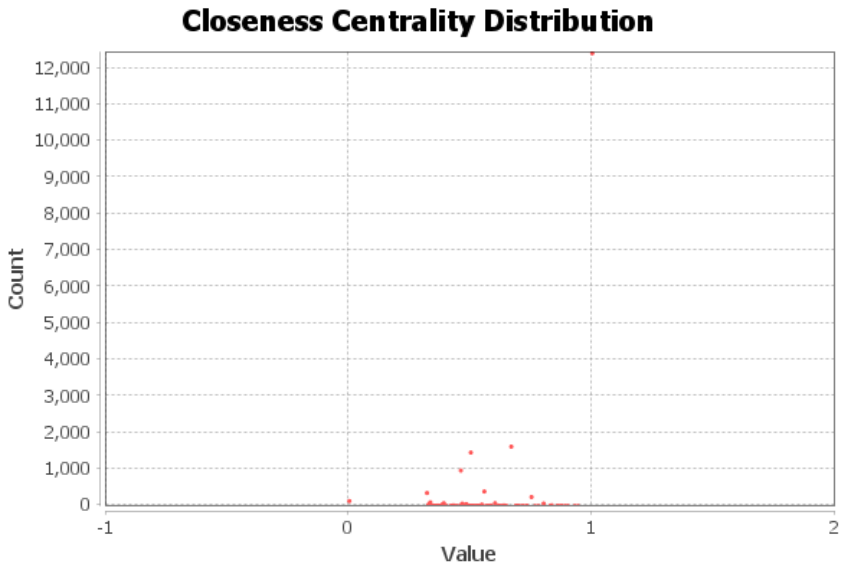




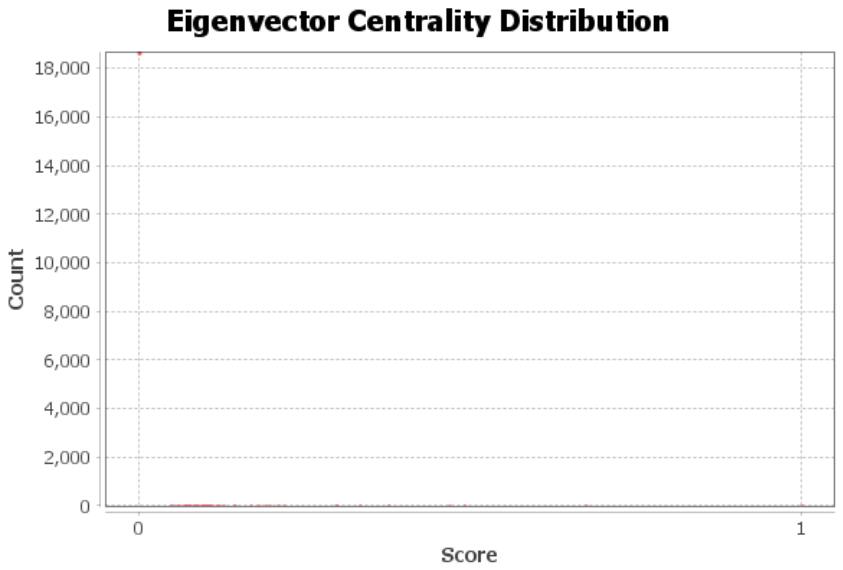
- **Betweenness Centrality**: This metric reveals nodes that act as critical intermediaries or "bridges" within the network. The distribution is highly skewed towards zero, showing that most nodes do not significantly control the flow of information. However, a few nodes exhibit high betweenness centrality, underscoring their importance in linking otherwise disconnected subgroups.



- **Closeness Centrality:** Closeness centrality evaluates how efficiently a node can reach all other nodes in the network. The relatively short average path length of 2.154 highlights a compact structure, where most nodes can access others through a small number of steps.



- **Eigenvector Centrality:** This measure emphasizes nodes connected to other influential nodes. While most nodes exhibit low eigenvector centrality scores, a select few stand out, reflecting their pivotal roles in maintaining the network's connectivity and influence hierarchy.



Gephi visualizations provided a detailed representation of the centrality metrics. Figure 4 focuses on degree centrality, using node size to emphasize the most connected nodes. In Figure 5, a comparative visualization of closeness, degree, eigenvector, and betweenness centralities reveals the hierarchical structure and key influential nodes. These visualizations were created using hierarchical layouts, underscoring the network's directed nature and strategic nodes.

These findings are integral to understanding the network's underlying structure, particularly in identifying key nodes that facilitate efficient information dissemination and network stability. The visualizations further highlight the interconnectedness and resilience of the BlueSky social network, offering actionable insights for strategic interventions.

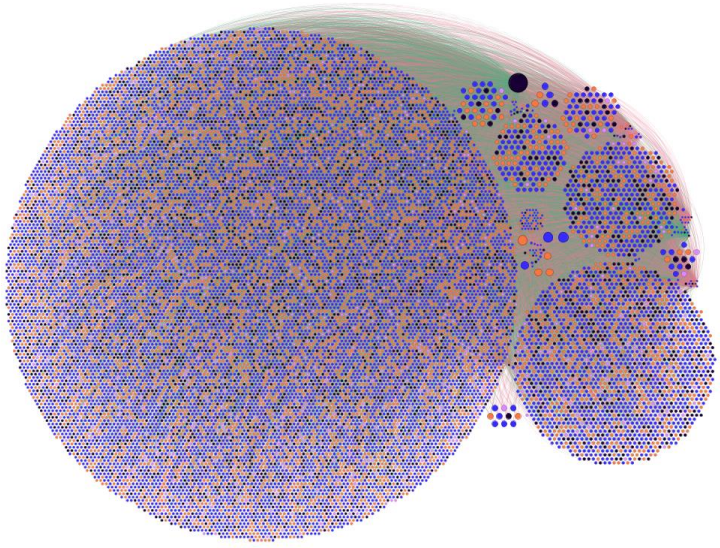


Figure 4: Visualization of degree centrality, highlighting the most connected nodes in the network

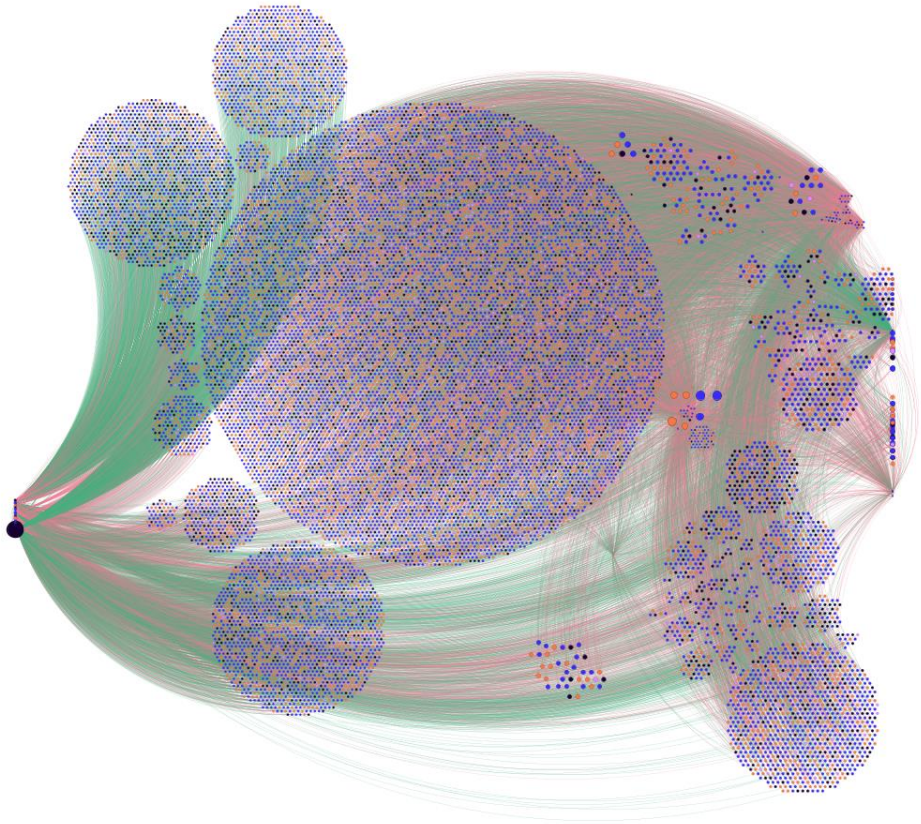


Figure 5: Comparative visualization of closeness, degree, eigenvector, and betweenness centralities, emphasizing node influence and connectivity

8. Clustering Effects in the Network

The analysis of clustering effects sheds light on the local connectivity of nodes within their immediate neighborhoods, providing insights into the density of connections and the tendency for triadic closures. Metrics such as the average clustering coefficient, number of triangles, and the distribution of clustering coefficients were examined to evaluate these characteristics.

The average clustering coefficient was calculated as 0.000755786, indicating sparse local connectivity across most regions of the network. The number of triangles, a measure of localized clusters, ranged from 0 to 8, highlighting areas of dense interactions. These triangles demonstrate the presence of triadic closures, where nodes form tightly connected subgroups, often indicative of strong local relationships.

For visualization, nodes were partitioned based on their clustering coefficients, with node sizes scaled proportionally to triangle counts. Larger nodes represent higher triangle

counts, effectively emphasizing regions with denser connectivity. This method highlights localized clusters and areas of frequent interaction.

As shown in Figure 6, the distribution of clustering coefficients and the visualization of triangle counts provide a comprehensive view of the network's local connectivity patterns. These insights illustrate the network's structure, capturing both sparsely connected regions and densely interconnected subgroups, which often mirror friendship-like patterns within the network.

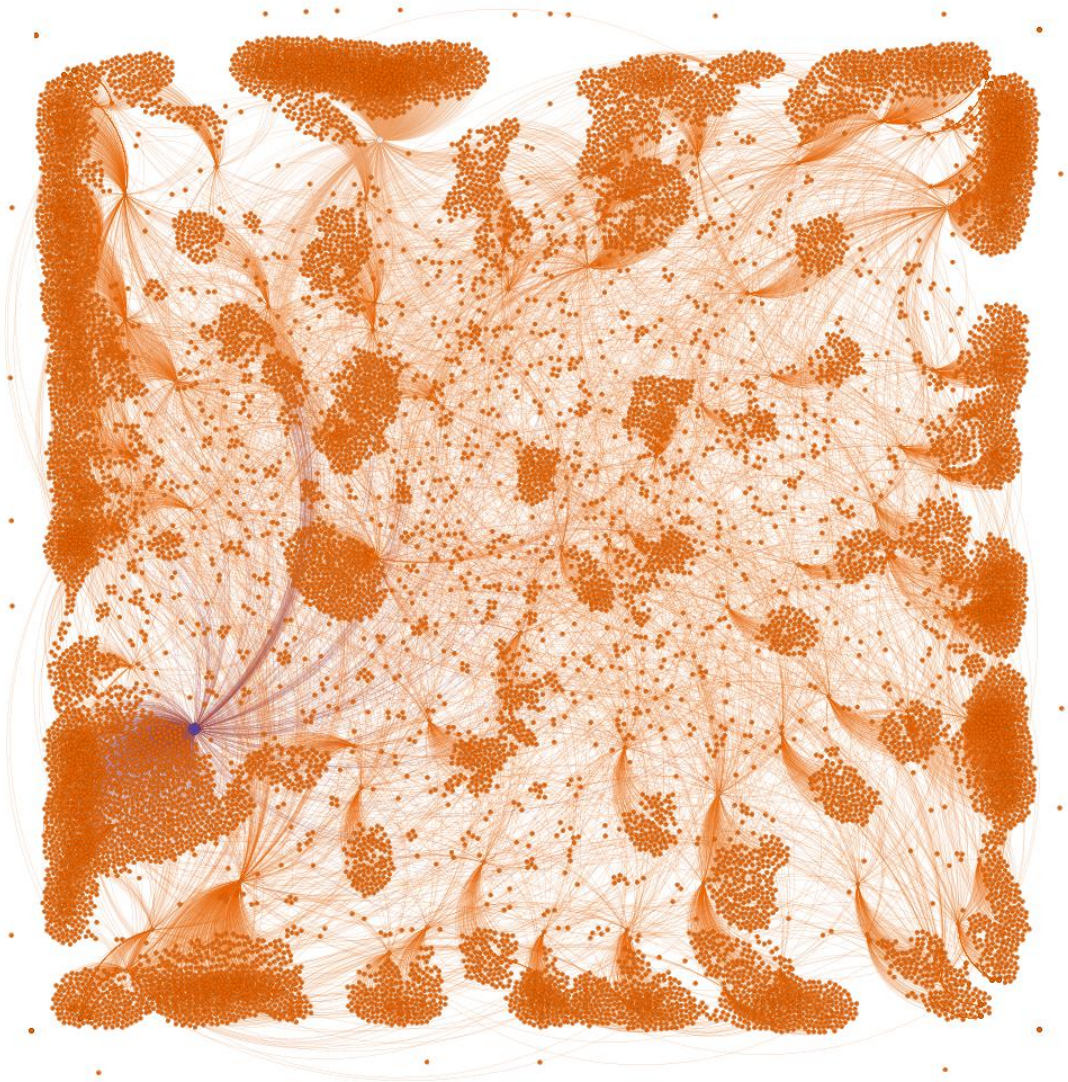


Figure 6: Visualization of clustering effects, showcasing local connectivity and dense regions

9. Bridges

The analysis of bridges within the network focuses on identifying edges that serve as critical connections between clusters. Local bridges, in particular, are edges that connect

nodes with minimal shared neighbors, acting as unique conduits for connectivity between otherwise disconnected regions. These edges, if present, play a crucial role in maintaining the structural integrity of the network by serving as bottlenecks or pathways.

Upon analysis, the results indicate that no edges qualify as local bridges, with all edges marked as FALSE. This finding highlights the network's robust interconnectedness, where multiple pathways exist between nodes, eliminating dependency on any single edge to maintain overall connectivity. The absence of local bridges suggests a highly resilient structure, capable of withstanding potential disruptions without significant fragmentation.

For visualization, edges were partitioned based on the IsLocalBridge attribute, with blue representing FALSE. Since no TRUE local bridges were identified, all edges appear uniformly blue. To further enrich the analysis, a filter was applied to edges using betweenness centrality, emphasizing edges with higher significance in connectivity pathways. These high-betweenness edges highlight potential critical conduits for information flow within the network.

Nodes in the visualization were ranked by degree centrality, with size reflecting the number of direct connections each node possesses. This approach underscores the influence of highly connected nodes in maintaining the network's structure and interaction dynamics.

The resulting visualization, depicted in Figure 7, effectively illustrates the network's resilience. It highlights the even distribution of connectivity across clusters and demonstrates the absence of single-edge dependencies, reinforcing the stability and robustness of the network's structure.

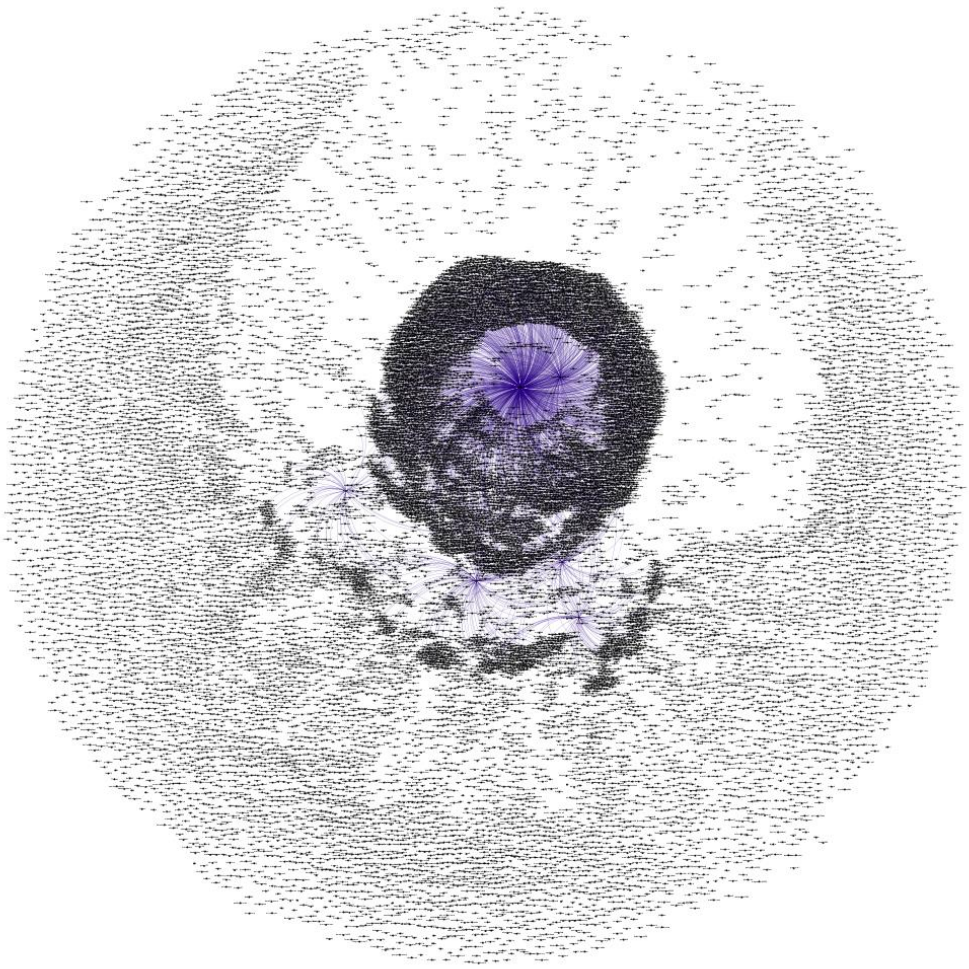


Figure 7: Visualization of bridges, emphasizing network resilience and critical pathways

10. Gender and Homophily Analysis

The analysis of gender and homophily within the BlueSky network offers valuable insights into social dynamics and interaction patterns. Out of the total 22,414 edges in the network, 9,085 edges (40.53%) connect users of the same gender, indicating a moderate level of homophily. The calculated E-I Index of 0.19 suggests a slight preference for cross-gender interactions (heterophily), reflecting a balanced network where users engage across gender boundaries while forming distinct gender-based clusters.

To visualize these dynamics, two distinct approaches were employed:

1. **Simplified Gender Partition and Interaction Patterns (Figure 8)**
   A streamlined visualization was created to depict the distribution of genders and their interaction patterns. Distinct gender-based clusters are highlighted, with cross-gender edges emphasizing the network's balance between homophily and heterophily. This

representation effectively captures the interaction dynamics while providing a high-level overview of the network's gender structure.

2. **Gender-Based Clustering with Hierarchical Attributes (Figure 9):** A circle pack layout was employed to illustrate the hierarchical structure of the network. Nodes were partitioned by gender, with blue for males, pink for females, and black for users with unknown gender. Additional attributes such as hub scores, modularity classes, triangle counts, and degrees were layered to enrich the visualization. This detailed perspective highlights the roles and connectivity patterns within gender-based clusters, emphasizing the interplay between hierarchical factors and network dynamics.

The visualizations (Figures 8 and 9) provide complementary perspectives. Figure 8 offers a broad overview of gender-based clustering and interaction patterns, while Figure 9 delves into the roles and hierarchical attributes within these clusters. Together, they reveal a nuanced balance of diversity and clustering within the network, underscoring the interplay of homophily and heterophily in shaping user interactions.

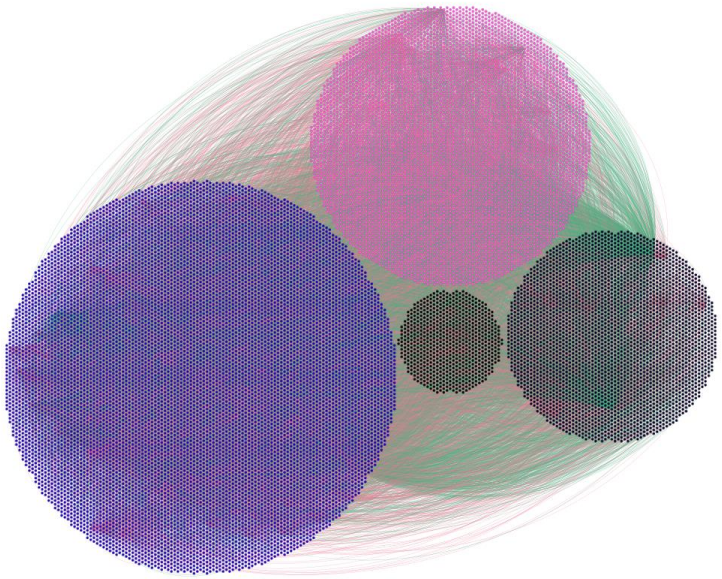


Figure 8: Simplified gender partition showing interaction patterns and cross-gender connections

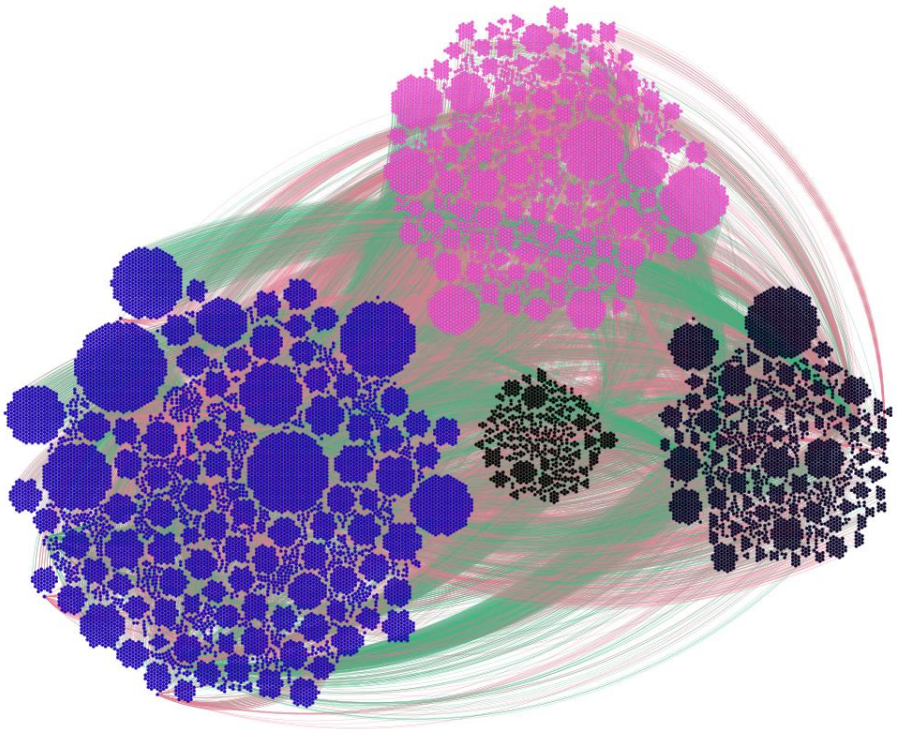


Figure 9: Gender-based clusters visualized with hierarchical attributes and connectivity roles

11. Graph Density

The Graph Density Report provides insights into the overall connectivity of the network by measuring the proportion of possible connections that are realized. Graph density is a fundamental metric in network analysis, calculated as:

$$\text{Density} = \frac{E}{N\,.\,(N-1)}$$

Where:

- E: Number of edges in the network
- N: Number of nodes in the network

For directed graphs, the formula considers all possible directed connections between nodes.

Parameters:

- **Network Interpretation:** Directed

Results:

- **Density:** 0.000

This extremely low density indicates a sparse network, with only a negligible fraction of possible connections realized. Such a result is typical in large social networks, where interactions are concentrated within specific clusters or groups rather than uniformly distributed across the network.

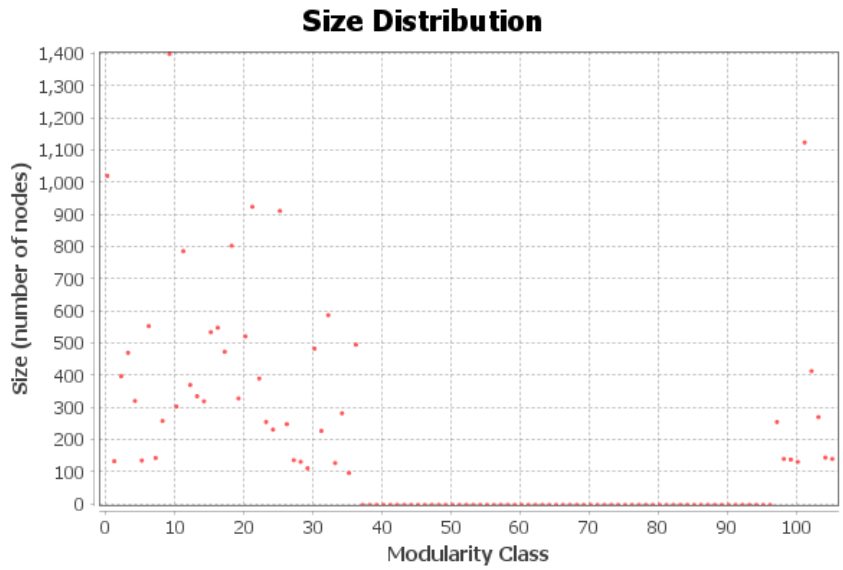
12. Community Structure Analysis (Modularity)

The modularity analysis provides critical insights into the community structure of the BlueSky network, showcasing how nodes cluster into tightly connected groups. Modularity, a measure of the strength of community division, was calculated at two different resolutions, offering a multi-level perspective on the network's organization.

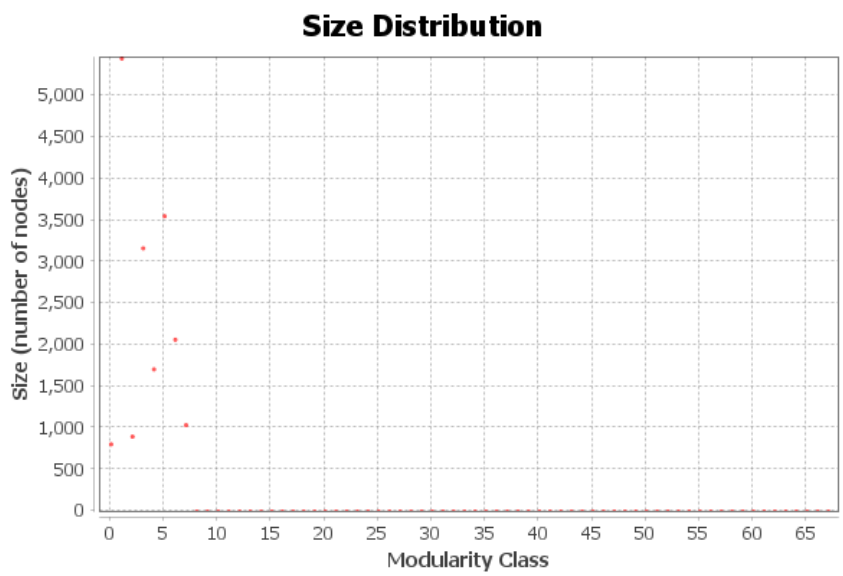
At a resolution of 1.0, the modularity score was 0.823, identifying 104 distinct communities. Increasing the resolution to 5.0 resulted in a modularity score of 0.727, with the number of communities reducing to 68. These results highlight the network's strong modularity, where nodes within communities are densely interconnected, while inter-community connections remain sparse.

Modularity Class Size Distribution:

- **Resolution 1.0**: The network is divided into 104 communities with sizes ranging from large clusters containing over 1,000 nodes to smaller groups with fewer than 100 nodes. This distribution emphasizes a decentralized structure where interactions are concentrated within distinct clusters.



- **Resolution 5.0**: At this higher resolution, smaller clusters merge into broader communities, reducing the number of groups to 68. This consolidation reveals larger interaction hubs, reflecting overarching structural patterns in the network.



At a lower resolution (Figure 10), the network's modularity structure uncovers distinct community boundaries, where nodes are tightly connected within their groups but sparsely connected across different groups. This fine-grained visualization emphasizes localized interaction patterns.

In contrast, at a higher resolution (Figure 11), the network consolidates into broader clusters, offering a higher-level abstraction of its structure. This perspective highlights the hierarchical organization of the network, where smaller subgroups coalesce into larger interaction hubs, providing a macroscopic view of community dynamics.

The findings underscore a well-organized and modular network, where users within the same community exhibit strong internal connections, likely reflecting shared interaction

patterns or interests. The dual-resolution approach illustrates both granular subgrouping and overarching structural hierarchies, offering valuable insights into the network's dynamics, potential pathways for information dissemination, and opportunities for targeted engagement strategies.

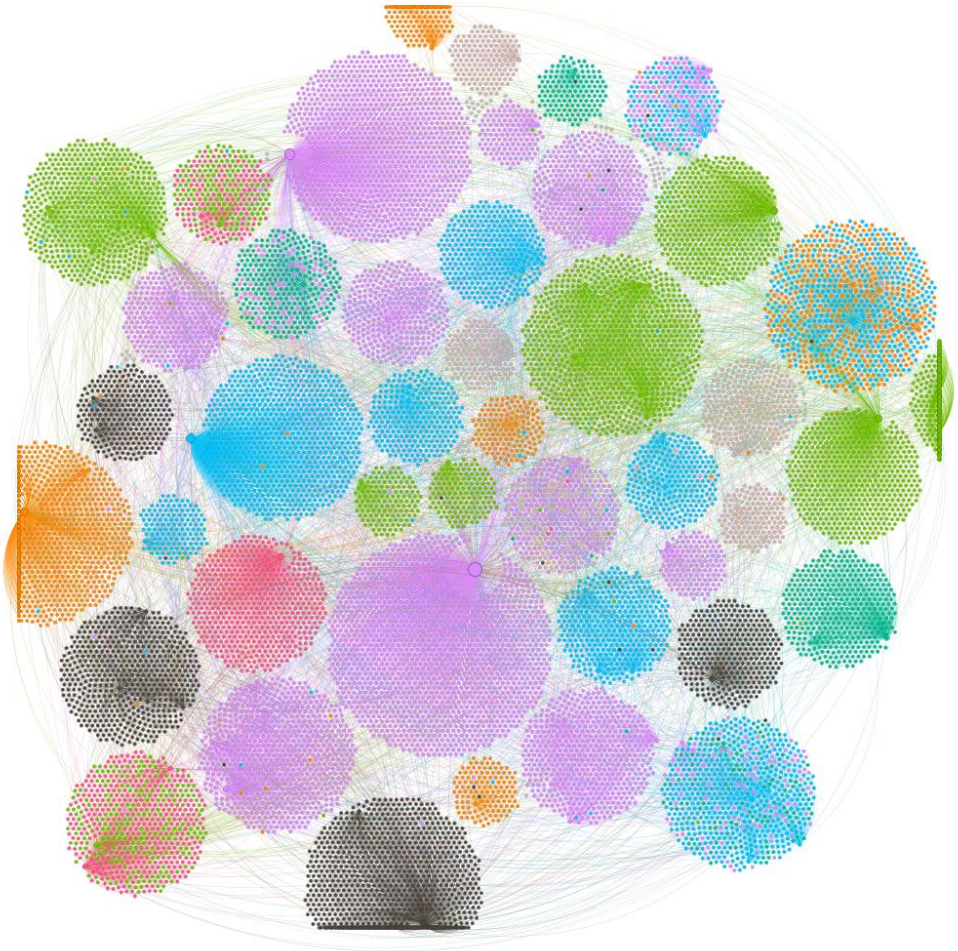


Figure 10: Visualization of 104 community structures at Resolution 1.0.

Figure 11: Visualization of 68 community structures at Resolution 5.0.

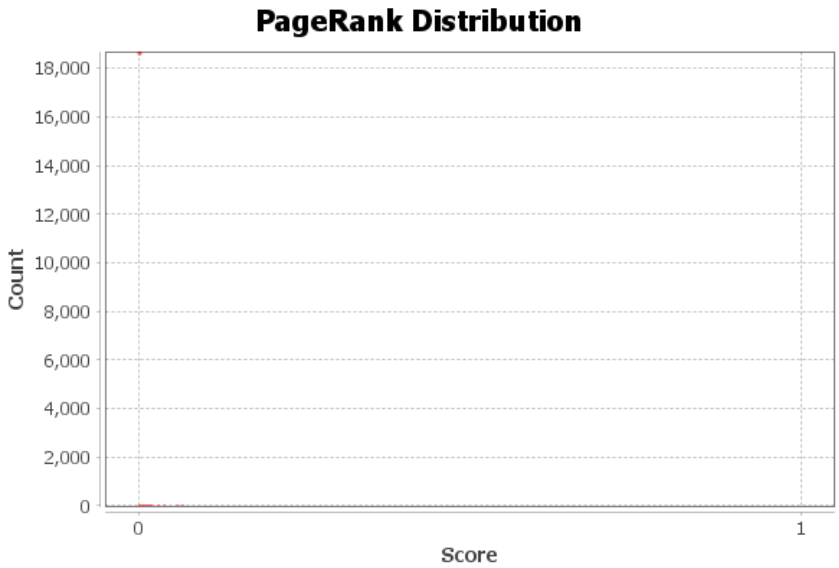
13. PageRank

The PageRank analysis underscores the influence of nodes within the network, identifying key individuals who hold significant centrality based on their connections. This metric evaluates the importance of nodes by considering both the quality and quantity of their connections, emphasizing the role of highly connected nodes in shaping the network's structure and information flow.

Using an Epsilon value of 0.001 and a Probability (α) of 0.85, the PageRank algorithm iteratively calculates scores to determine the relative importance of each node. The results reveal a highly uneven distribution of influence, with a small subset of nodes exhibiting disproportionately high PageRank scores. This distribution highlights the "rich-get-richer" phenomenon, a common feature in social networks where a few nodes serve as influential hubs.

Key findings from the PageRank distribution:

- The distribution highlights that while most nodes hold low scores, a select few dominate with significantly higher PageRank scores. This reflects a hierarchical network where influence is concentrated among top nodes, which serve as primary hubs for connectivity and information flow.



The visualization (Figure 12) adopts a circular layout, with node sizes scaled proportionally to their PageRank scores. Larger nodes represent higher influence, and key individuals are prominently labeled to highlight their importance. The layout reveals a dense core of highly influential nodes surrounded by less-connected peripheral nodes, emphasizing the hierarchical structure of the network.

These findings are crucial for understanding the flow of information and influence within the network. By identifying top influencers, targeted strategies can be developed to enhance information dissemination or engagement efforts. The combined visualizations offer a comprehensive view of the network's power dynamics, emphasizing the critical roles played by highly connected and influential nodes.

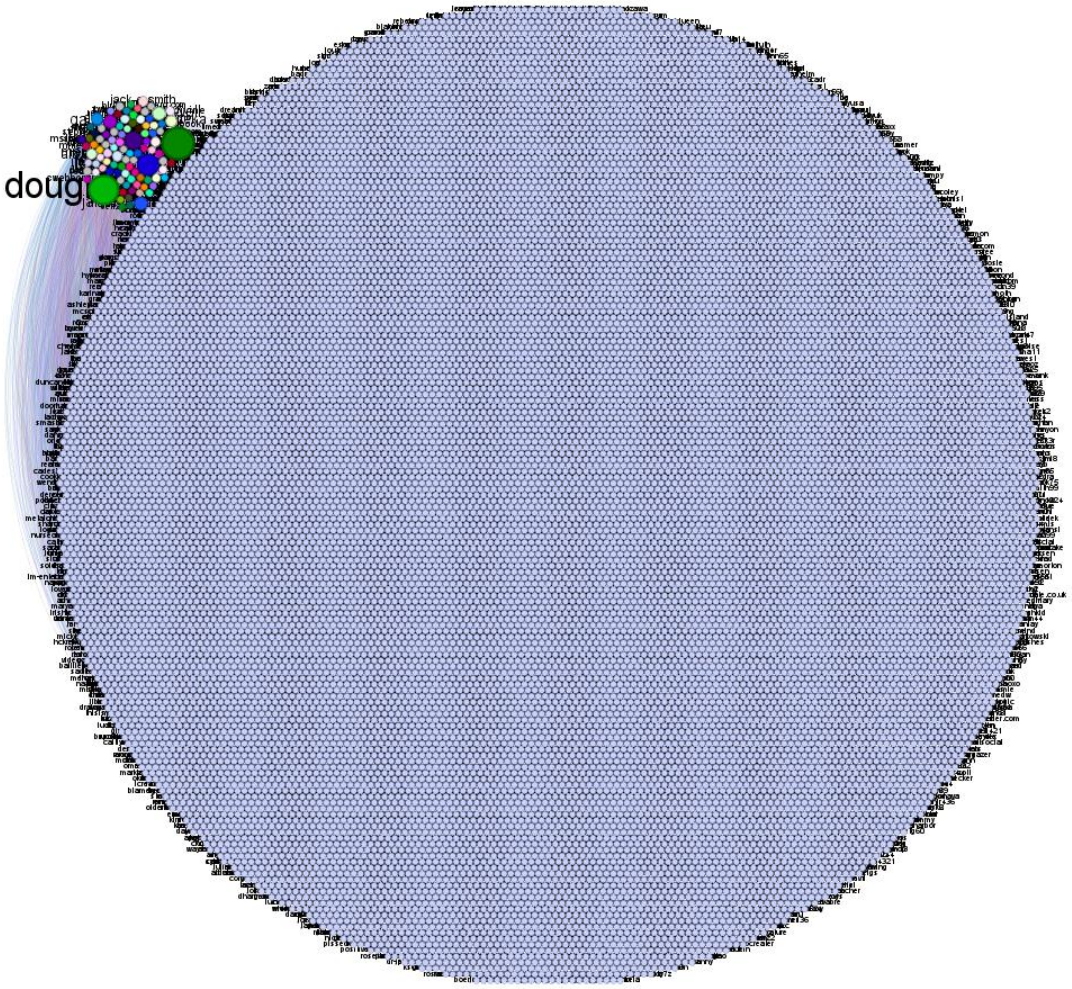


Figure 12: Circular visualization emphasizing nodes with high PageRank scores

Conclusion

In conclusion, the analysis of the BlueSky social network unveils several critical insights into its structure and dynamics. The network exhibits a small-world topology, characterized by tightly connected central components surrounded by sparsely linked peripheral nodes. Centrality metrics identify key influencers who act as bridges and hubs, playing a pivotal role in ensuring connectivity and facilitating the flow of information across the network.

The gender-based analysis highlights a moderate level of homophily, with cross-gender interactions contributing to the network's balance and diversity. Community structure analysis reveals well-defined clusters that reflect localized interactions, which could serve as strategic focal points for targeted engagement efforts.

Overall, this study offers a comprehensive understanding of the network's connectivity, influence distribution, and community dynamics. These findings provide a strong foundation for further exploration of social influence, information dissemination, and strategic community-building efforts within the BlueSky platform.